

A Current Reuse CMOS RF Front-End for 4G LTE Systems

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Abstract

This proposed architecture is a single-end to dual-end CMOS RF front-end which is applied for LTE system. This work use Digital IF receiver architecture including current reuse source inductive degeneration LNA and folded-cascode mixer in accord with 3GPP LTE system standard. Since LTE is a communication protocol applied on handheld devices, the demands of circuit designs are based on low power and high gain. The most important parameters of front-end circuits of receivers are Noise Figure and linearity. The central frequency of RF circuit operates between 2360 to 3010MHz including LTE stand band 7. The front-end achieves 24.032 dB conversion gain, 6.48 dB NF, -14 dBm IIP3, -23.5 dBm P1dB and 2.532 mW power consumption. All the circuits are designed in 0.18 μ m CMOS process with 1.2V supply voltage.

Keywords: front-end receiver, long term evolution (LTE), low noise amplify (LNA), mixer, low power consumption

1. Introduction

Long Term Evolution (LTE) is the next generation of 4G mobile wireless broadband communication system. The utilized band is mainly allocated based on Evolved Universal Terrestrial Radio Access (E-UTRA) which is part of 3GPP LTE air interface. E-UTRA is a whole new system belonging to 3GPP 8 editions not incompatible with wideband code division multiple access (W-CDMA). For data transfer, download uses Orthogonal Frequency Division Multiple Access (OFDMA), and upload uses Single-carrier Frequency-Division Multiple Access (SC-FDMA) [1-3]. Operating bands of proposed architecture is from 2620 to 2690MHz within Band7 of E-UTRA. Flexible bandwidths are 5MHz, 10MHz, 15MHz and 20MHz. Transferring speed based on 20MHz frequency can reach to 100Mbps for downlink and 50Mbps for uplink. Chip is designed in 0.18 μ m CMOS process.

Communication IC is target to a small size and low power circuit in the future. Along with increasing of mixed circuits, reliability requirements of CMOS become more important. Front-end circuit of RF is the most popular research topics in recent years. In sum, the target of CMOS communication circuit's integration is to build a RF front-end and base-band processing receivers [4-6].

This paper is organized as follows. Section II presents the RF front-end receiver system architecture for LTE. Section III describes the proposed hardware design. Section IV shows the simulation results, and Section V concludes the paper.

2. Systems Architecture

The LTE supports both frequency-division duplex (FDD) and time-division duplex (TDD). From 700MHz to 5.8GHz transmission band, the smallest bandwidths of signal is from 1.4MHz to 20MHz. Fig. 1 is the receiving architecture of digital IF(Intermediate frequency). The advantage of this architecture is the mixer without channel I/Q. It not only avoids phase and gain imbalance, but also prevents DC offset of direct conversion receiver and flicker noise.

This paper focuses on LNA (Low Noise Amplifier) and mixer of RF Front-End. To meet the design requirement, it needs to make a balance between noise figure and linearity. This proposed architecture includes a single-end source inductive degeneration LNA and folded-cascode mixer. And also it must meet the requirements of common design features for the receiver front-end like Noise Figure(NF), input-referred third-order intercept point (IIP3) and input power 1-dB compression point (P1dB), and Conversion Gain. Low power consumption as well as good noise performance is a great challenge to design.

Table I describes the receiver specification of LTE including the input power, the noise figure, input third-intercept point, and input 1dB compression point. The requirements can be calculated from the 3GPP TS 36.101 V8.9.0 standard's specification [7].

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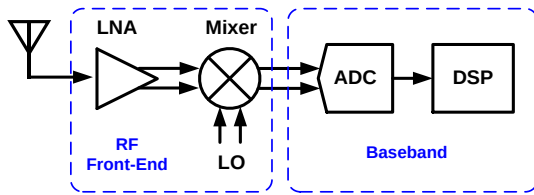


Figure 1: Digital IF receiver architecture

Table 1: LTE receiver specifications

Receiver Items	Specification
Input Power [dBm]	-95~-25 dBm
Noise Figure [dB]	<7
P1dB	>-25
IIP3 [dBm]	>-15

3. Circuit Design

3.1 Receiver RF Front-End

There are 2 compositions of Receiver RF front-end including LNA and Mixer cascaded configuration in Fig 2. The benefit of this architecture has better conversion gain overall circuit. Moreover, higher gain means better noise suppression. However, the power consumption is higher because of higher currents from 3 biases voltage including I_{LNA} , I_{mixer1} , and I_{mixer2} . This design is not suitable for a low power application especially in series schema. Another design includes LNA and Mixer cascoded configuration in Fig 3. The way to connect with LNA and mixer DC path is called current reused. Comparing with cascaded structures, there are lower power consumption but higher noise figures in cascade. The input signal directly goes to switch stage of mixer without amplified twice; the gain is smaller in in cascoded structure. And higher noise figure is caused by bigger current in switch stage of mixer. Also lower gain means worse noise suppression. The proposed design for RF front-end of

a receiver is shows in Fig. 4 including merged LNA and folded-cascode mixer. Stacked single-ended LNA and double balanced Mixer are included in internal configuration. The switch stage of Mixer has smaller dc current than LNA. Comparing with standard cascade, folded-cascode has same gain but less power consumption.

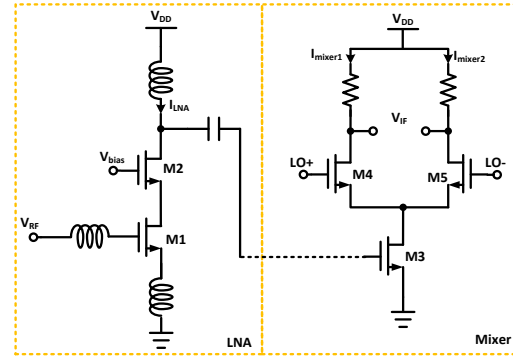


Figure 2: LNA cascade mixer

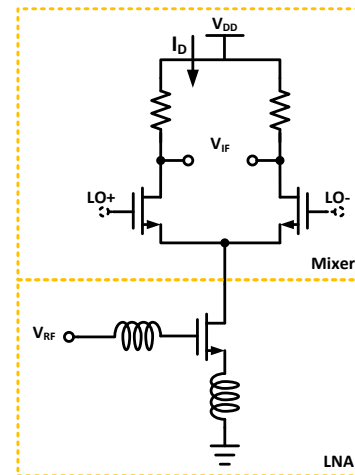


Figure 3: LNA cascode mixer

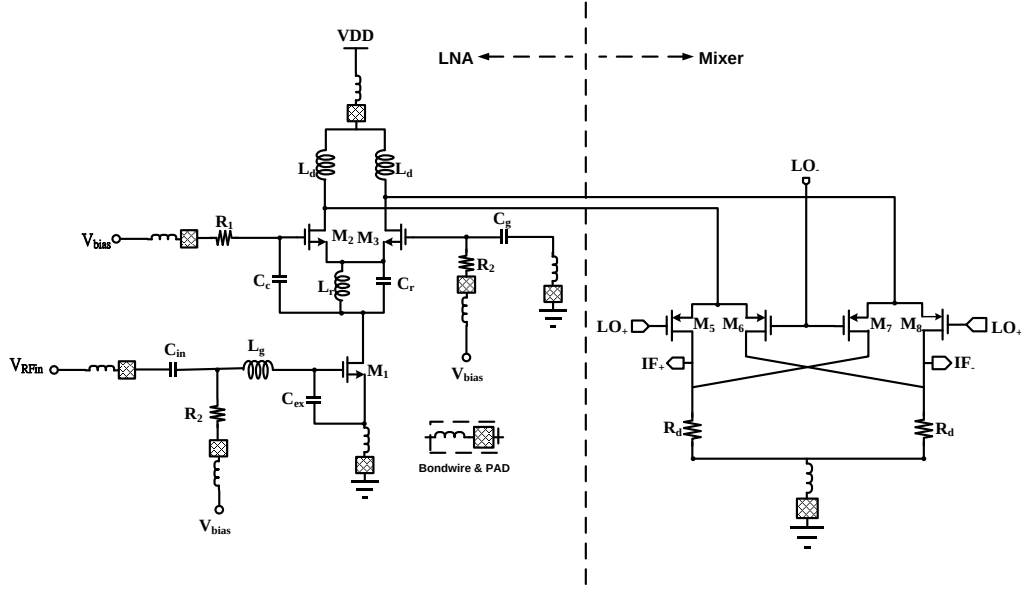


Figure 4: Proposed folded-cascode receiver RF front-end with merged LNA and mixer

3.2 Low Noise Amplifier

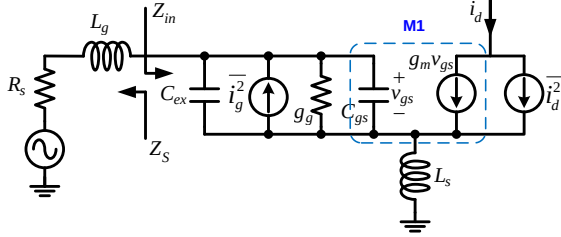


Figure 5: Small signal equivalent circuit of inductive source degeneration low noise amplifier

M₁ of Fig. 4 is common-source LNA with inductive source degeneration. Part of source L_s is replaced by bond wires to decrease size of a chipset. Besides, extra capacitance C_{ex} is added on M₁ source and Gate to increase degree of freedom for input noise matching. Furthermore, resonant circuit on M₁ Drain between L_r and C_r set to 2.65GHz is to let RF signal amplified twice by going through C_c capacitor to M₂ common-source amplifier for a higher gain. The following are the formulas of input resistant Z_{in}. According to KVL and KCL law, t V and I are derived:

$$\begin{cases} Z_{in} = \frac{v}{i} \\ v = i \cdot sL_g + \frac{i}{s(C_{gs} + C_{ex})} + (i + g_m V_{gs})sL_s \\ i = V_{gs} \cdot sC_{gs} \end{cases} \quad (1)$$

$$g_m V_{gs} = g_m \frac{i}{sC_{gs}} \quad (2)$$

Multiplying sL_s with item $(i + g_m V_{gs})$, we can get the following Equation (3) from Equation (2)

$$v = i \cdot sL_g + \frac{i}{s(C_{gs} + C_{ex})} + i \cdot sL_s + \frac{g_m L_s}{C_{gs}} \cdot i \quad (3)$$

$$= i \left(s(L_g + L_s) + \frac{1}{s(C_{gs} + C_{ex})} + \frac{g_m L_s}{C_{gs}} \right)$$

The input impedance of the LNA is given by

$$Z_{in} = \frac{v}{i} = s(L_g + L_s) + \frac{1}{sC_t} + \frac{g_m L_s}{C_{gs}} \quad (4)$$

Where $C_t = C_{gs} + C_{ex}$. From Equation (4), it shows that L_s helps to generate a real part for input impedance. For measurement consideration, $L_s(g_m/C_{gs})$ should be set equal to 50 ohm, and imaginary term should be zero at operating frequency. Therefore, we can let the input impedance as R_{in} . Fig. 5 shows the small signal equivalent circuit of LNA, where V_{in} and I_{in} are input voltage and current. The V_{gs} can be presented as

$$V_{gs} = \frac{I_{in}}{\omega_0 C_t} = \frac{V_{in}}{R_{in} \omega_0 C_t} \quad (5)$$

and the G_m of the LNA is given by

$$G_m = \frac{I_{out}}{V_{in}} = \frac{g_m V_{gs}}{V_{in}} = \frac{g_m}{R_{in} \omega_0 C_t} \quad (6)$$

Where ω_0 is operation frequency. Thus, the power across the load can be written as

$$P_0 = \frac{V_{out}^2}{R_{load}} = \frac{[I_{out} \cdot (R_{load} // R_{out}) V_{gs}]^2}{R_{load}} = \frac{g_m^2}{R_{in} \omega_0^2 C_t^2} \quad (7)$$

The power gain is approximately presented as

$$G_c = \frac{P_o}{P_i} = \left[G_m \frac{R_{out}}{R_{load} + R_{out}} \right]^2 \cdot R_{load} \quad (8)$$

3.3 Folded-cascode Mixers

On the basis of formulas proposed by Friis [13], noise could be lower, and gain could be higher for increasing stages of series system. In the receiver system, the noise made by Mixer could be suppressed by high gain LNA. For analysing flicker noise inside Mixer, we divided Mixer to 3 components including RF input gain stage, LO switching stage and IF load. Poly-silicon resistors are used for IF load with less flicker noise. Gain stage input and flicker noise of RF input signal are frequency shifting in a switch stage. Based on the principle of Mixer, flicker noise will be transferred to LO frequency or to other band whose central frequency is odd. And it won't be shown on a baseband. Flicker noise of the whole system is caused by LO switching stage in Mixer [8-9]. The flicker noise current is expressed as

$$\bar{i}_n^2 = \frac{K}{f} \cdot \frac{g_m^2}{WLC_{ox}^2} \cdot \Delta f \approx \frac{K}{f} \cdot \omega_T^2 \cdot A \cdot \Delta f \quad (9)$$

K is process parameter and $A (=W \cdot L)$ is Gate size. From Equation (9), flicker noise, A and K are proportionally related. In PMOS, K value is always $10^{-28} C^2/m^2$, and K is 50 times smaller than NMOS transistor. Therefore, Flicker noise of PMOS is less than NMOS. For this purpose, we choose PMOS to make Mixer LO signal switch stage and lower down the effect of flicker noises [14].

In addition, when local oscillation signals are input to Mixer, noise impulses will be generated on a switch stage. The flicker noise current is expressed in the following formula:

$$i_{o,n} = \frac{(4I_{sw} \times V_n(f))}{(S_{LO} \times T_{LO})} \propto I_{sw} \quad (10)$$

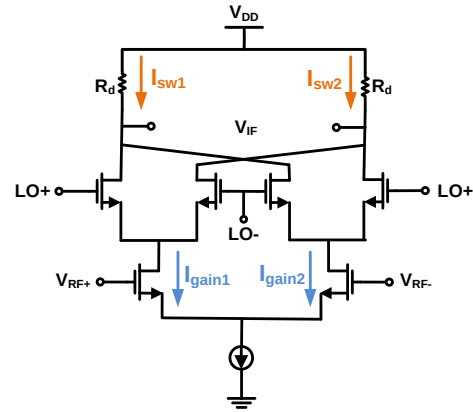
I_{sw} : Mixer switch stage bias voltage current.

V_n : Switch stage Equivalent of flick noise voltage

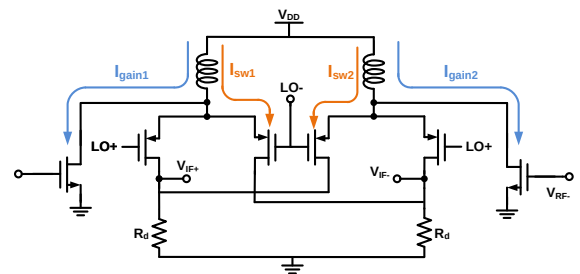
T_{LO} : Cycle Time of Signal LO

S_{LO} : Slope of signal LO

From Equation (10), spectrum of noise current on an input stage is proportional to bias voltage current of a switch stage. Therefore, lowering down I_{sw} is decreasing flicker noise affection. Fig. 6(a) shows traditional double-balance Mixer. Because of switch stage directly cascaded on gain stage, in order to get same gain, switch stage must use same bias voltage current with gain stage, which makes flick noise higher. The proposed architecture is why we use a folded-mixer. The purpose of this architecture shown on Fig. 6(b) is to divide operating bias voltage current between gain and switch stage to achieve high gain and low noise. Comparing with big bias voltage current of gain stage, switch stage only needs smaller bias voltage current to control. Power consumption of the whole system is not to increase, and flick noise is very low.



(a) Double-balance mixer circuit



(b) Folded-cascode mixer circuit

Figure 6: Double-balance mixer and Folded-cascode mixer circuit

4. Simulation Results

Fig. 7 is the layout of RF front-end which is designed in TSMC 0.18 μ m 1-poly six-metal CMOS process. The size of chipset is 1.2×1.2 mm². Fig.8 is the simulation result of input return loss (S_{11}). Gain of S_{11} for Central frequency 2.65GHz is -27.533dB, and the bandwidths is about 2.4GHz to 3GHz. Fig. 9 is the simulation results of P_{1dB} and conversion gain. P_{1dB} is -23.5dBm. While input power is -40dBm, conversion gain is 25dB. The intermodulation distortion is simulated by two-tone test, the 3rd-order input intercept point is about -14 dBm, and the 3rd-order output intercept point is about 1 dBm as shown in Fig.10. Fig.11 is the simulation results for Noise figure of Mixer. Mixer includes single-side band (SSB) and double side band (DSB). Generally, Noise Figure uses DSB for standard value NF_{dsb} which is 6.5dB at 10MHz. Fig. 12 is the Stability Factor of the whole system. K must be bigger than one which shows designed architecture close to stable [15].

The post-simulation results are shown on Table II and compared with other references. From table II, Conversion Gain is similar with other references. But power consumption is less. The distinguishing feature of proposed architectures uses lower power consumption to generate same gain. Table III is the comparison of pre-simulation and post-simulation. According to simulation results, only S_{11} becomes worse. The design meets specifications.

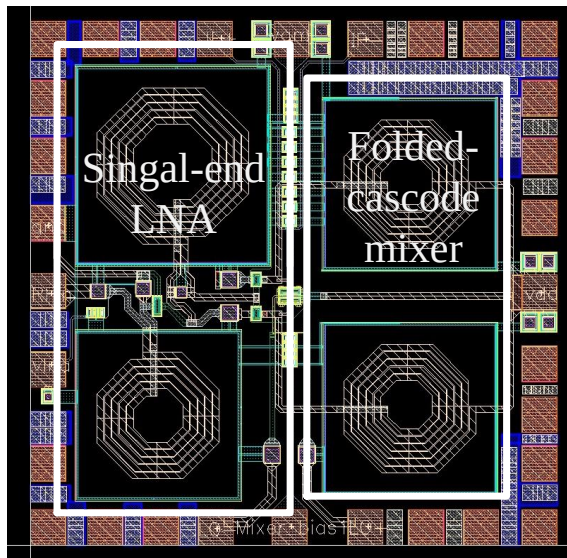


Figure 7: Chip layout of the proposed receiver RF front-end

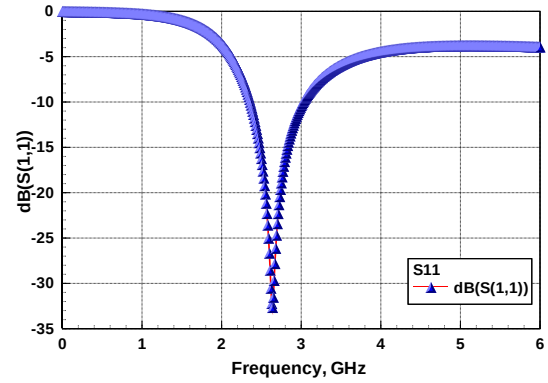


Figure 8: Input return loss

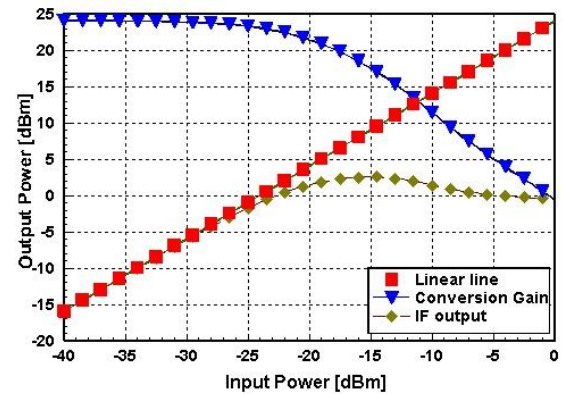


Figure 9: Simulation P_{1dB} and conversion gain plot

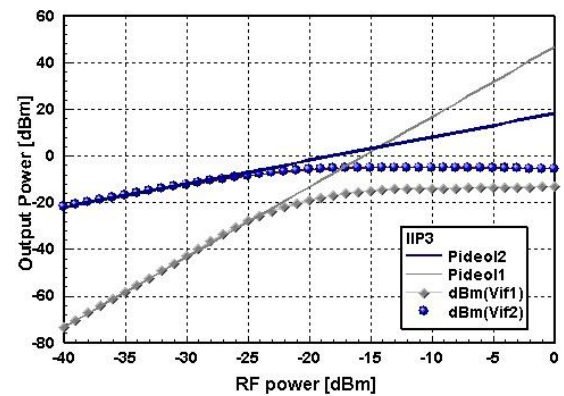


Figure 10: Simulation IP3

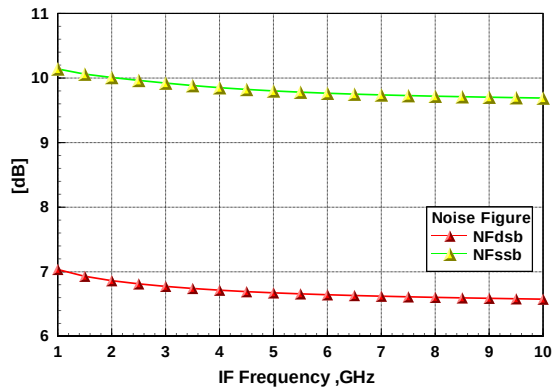


Figure 11: Simulation Noise figure

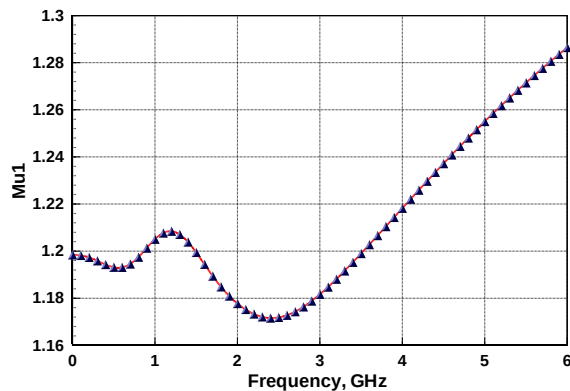


Figure 12: Stability Factor

Table 2: Summarized performances of proposed receiver and its comparison with previously published papers

Receiver Item	Post-simulation	[10]	[11]	[12]
V_{DD} [V]	1.2	1.8	1.8	1.8
Conversion Gain[dB]	24.03	28.69	38	25.5
Noise Figure [dB]	6.48	7.06	4-4.5	4.5
P_{1dB} [dBm]	-23.5	-19	-16	-20
IIP3[dBm]	-14	-9	-6	-11.2
Power Consumption [mW]	2.532	14.4	28	7.56
Frequency [GHz]	2.65	2.4	0.5~3	2.4
Process [μ m]	0.18	0.18	0.18	0.18

Table 3: Summarized performances of proposed receiver

Receiver Items	Pre-simulation Results	Post-simulation Results
VDD [V]	1.2	1.2
S11 [dB]	-44.043 @2.65 GHz	-27.559 @2.65 GHz
Voltage Conversion Gain [dB]	24.79	24.032
Noise Figure [dB]	6.42	6.48
P1dB [dBm]	-26	-23.5
IIP3 [dBm]	-16	-14
LO-RF Leackage [dB]	-85.054	-85.334
Power Comsumption [mW]	2.532	2.532

5. Conclusions

This paper proposed a receiver RF front-end which is applied in LTE system with current reused function. The architecture is implemented in TSMC 0.18 μ m CMOS process including single-end source inductive degeneration LNA, and folded-cascode mixer uses PMOS device to reduce the flicker noise. The receiver front-end provides 6.48dB NF, -14 dBm IIP3, -23.5 dBm P1dB, 24.032 dB conversion gain while consuming 2.11mA from a 1.2 V supply.

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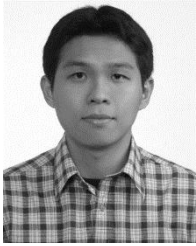
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